Ultrasonic Thickness Testing
Inspection Report Form

Report Number:	T-204	Date of Inspection:	07/12/2025

Job Detail			
Client Name:	INEOS Grangemouth	PO Number:	PO-4852
Job Description:	External and internal inspection with UT thickness readings of the floor plates		

Findings
The external and internal inspection of storage tank T-204 identified a number of age and service related deterioration mechanisms that require ongoing monitoring and localised remediation. External visual examination recorded several areas of coating breakdown on the lower shell courses, particularly around the 6 o'clock and 9 o'clock positions, with associated surface corrosion and local pitting. At the shell to bottom weld, light staining and minor undercut were observed at several locations, although no evidence of active product leakage was noted. Settlement readings taken around the circumference showed a maximum differential settlement of 8 mm compared with the previous survey, remaining within typical industry tolerance but trending upwards at one datum point, which should be trended at the next inspection. Internal inspection following gas freeing and cleaning showed isolated but distinct pitting on the floor plates within 500 mm of the shell, most pronounced near the product inlet. Ultrasonic thickness measurements in these areas indicated local wall loss of up to 30 per cent relative to nominal plate thickness, with minimum measured values of 6.3 mm against an original 9.5 mm plate. Pit depths of up to 2.0 mm were recorded, with a generally tight pit density rather than widespread general thinning. No through plate penetration was identified, and the shell courses remained above minimum recommended thickness, although one band of readings on the first course showed an emerging pattern of thinning along a weld seam. Nozzle and manway welds were examined by magnetic particle testing and were found to be free from reportable surface breaking indications. Roof structure, support beams and handrailing exhibited only minor coating breakdown and light rust staining. Overall, the tank remains fit for continued service in the short to medium term, provided that local floor repairs are carried out in the most affected zones and that corrosion trends at the shell to bottom region are closely monitored at the next planned inspection.

Applus UK Limited, Unit 2, Blocks C & D, West Mains Industrial Estate
Falkirk, FK3 8YE

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Photographs	
Images	**Description**
	Tank plate
	Outside of tank

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Photographs	
Images	**Description**
	Stairs
	Valves

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